Universidad Distrital Francisco José de Caldas
School of Engineering – Computer Engineering Program

Data Analysis Programming – Semester 2026-II

Workshop No. 1

Hazard Definition, Source Validation, and Data Contract

Team 8 – Tsunami Monitoring

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Executive Summary

[TODO: Five to eight lines: hazard, scope, the four sources, main findings of the profiling, and the proposed indicator.]

1 Hazard Definition

1.1 Phenomenon and Severity Measure

A tsunami is a series of long ocean waves generated by the sudden displacement of a large volume of water. Most tsunamis are caused by large undersea earthquakes, but landslides, volcanic eruptions and meteorological events can also generate them. In our HazEL sample, 36 of 221 events (16%) have a non-earthquake cause (landslide 17, meteorological 9, volcano 6, unknown 2, among others).

The canonical severity of the system is the earthquake magnitude, stored together with its type (for example mww, mwc, mb or Ml), because magnitude types differ across sources. Depth, in kilometers, is kept as a hazard-specific attribute. Since no single measure describes tsunami severity, the system combines the variables in Table 1.

Measure	Unit	Source	Meaning
Earthquake magnitude (`mag`, `eqMagnitude`)	Dimensionless; the magnitude type is stored	USGS, HazEL, tsunami.gov	Size of the earthquake that may generate a tsunami
Earthquake depth (`depth`, `eqDepth`)	km (inferred for HazEL)	USGS, HazEL, tsunami.gov	Shallow events are more likely to displace the seafloor
Maximum water height (`maxWaterHeight`)	m	HazEL	Observed tsunami height (historical only)
Event validity (`eventValidity`)	Ordinal scale from -1 to 4	HazEL	Confidence that the event was a real tsunami, from erroneous or doubtful entry to definite tsunami
Official alert (CAP `severity`, `certainty`)	Categorical (e.g. Minor / Unlikely)	tsunami.gov	Assessment by the warning centers
`tsunami` flag	0 / 1	USGS real-time	A tsunami bulletin *may* exist; it does **not** mean that a tsunami occurred

Table 1: Severity-related variables used by the system.

Note on depth in HazEL. The HazEL Swagger defines `eqDepth` as an integer (int32) and does not state its unit. We interpret it as kilometers because it matches the USGS depth (km) for the matched events: in 25 of the 28 NOAA–USGS pairs of the sample (Section 4) the difference is 2 km or less. Being an integer, it loses the decimals that USGS provides.

1.2 Why Monitoring Matters

Tsunamis can reach coasts thousands of kilometers from their source and cause major loss of life and damage, as in the 2011 Tohoku event (M 9.1), which appears in our samples. Monitoring recent earthquakes together with the historical tsunami record and the official bulletins helps an analyst judge quickly how relevant a new earthquake may be. The system supports monitoring and correlation and does **not** predict tsunamis; the indicator must never be read as a warning, and official warnings come only from the tsunami warning centers.

1.3 Geographic and Temporal Scope

The project covers the Pacific basin, including Alaska and Hawaii, during 2010–2025, with earthquakes of magnitude $M \geq 6$. NOAA HazEL regions 80–89 and a USGS rectangle (latitude -60 to 65, longitude 100 to 300) provide geographic filters, and a Pacific polygon removes the non-Pacific parts of the rectangle (Section 2); the USGS historical sample covers 2010–2011 to meet the workshop's 200-record requirement. The two real-time feeds are profiled in full, with Pacific events identified by coordinates; tsunami.gov covers the Pacific, Alaska, the US and the Caribbean.

Source	Coverage	Fits the scope?
NOAA HazEL	Global	Yes, regions 80–89
USGS FDSN	Global	Yes, Pacific rectangle
USGS real-time feeds	Global	Yes, in-scope events identified by coordinates
tsunami.gov	Pacific, Alaska, US, Caribbean	Yes, in-scope events identified by coordinates

1.4 Meaning of "Current Situation"

For this system the current situation comprises:

- earthquakes with $M \geq 4.5$ inside the Pacific scope during the last 24 hours and the last 7 days (USGS real-time feed);
- the latest tsunami.gov message of each warning center (NTWC and PTWC), including information statements;
- earthquakes of the last 7 days flagged by USGS with `tsunami = 1`.

$M \geq 4.5$ is the minimum magnitude of the USGS "past month" feed that we use, and these earthquakes feed the recent-events list of the dashboard. In contrast, the statistics, the comparison with the historical record and the indicator use only earthquakes with $M \geq 6$, the threshold of the historical catalog, so that both sides of the comparison follow the same criterion and small earthquakes today do not look unusual only because the history does not contain them.

2 Historical Sources

2.1 NOAA/NCEI HazEL – Global Historical Tsunami Database

Owner	NOAA National Centers for Environmental Information (NCEI)
URL	https://www.ngdc.noaa.gov/hazel/view/swagger
Access method	REST API, JSON
Authentication	None required
Rate limits	Not documented. Team policy: identified User-Agent (`UDistrital-Team8-Tsunami-Monitoring/1.0`), three retries with increasing wait, sequential requests, and caching of every downloaded file.
License and attribution	Public use, no use constraints, according to the dataset metadata. Required citation: National Geophysical Data Center / World Data Service, *NCEI/WDS Global Historical Tsunami Database*, NOAA National Centers for Environmental Information, doi:10.7289/V5PN93H7, accessed 8 November 2026 [1].
Fields used	`id`, `year`, `month`, `day`, `hour`, `minute`, `second`, `latitude`, `longitude`, `causeCode`, `eventValidity`, `eqMagnitude`, `eqDepth`, `maxWaterHeight`, `warningStatusId`, `regionCode`, `numRunups`
Time coverage	The database goes back to 2100 BC; we use 2010–2025 (last event in the sample: 12 December 2025).
Update policy	Curated catalog; its metadata record gives the update frequency as "as needed". Records can be revised or reclassified (for example their validity).

Evidence. Query used to obtain page 1; the unmodified response is stored in `workshop_1/samples/noaa_tsunamis_pacific_2010_2025.json` (page 2 in `noaa_tsunamis_pacific_page2.json`, code tables in `noaa_descriptor_*.json`):

```
GET https://www.ngdc.noaa.gov/hazel/hazard-service/api/v1/tsunamis/events
    ?minYear=2010&maxYear=2025&itemsPerPage=200
    &regionCode=80,81,82,83,84,85,86,87,88,89&page=1
```

A short excerpt of the response is in Appendix A.

Volume and partitioning. The response reports `totalItems` = 221 over two pages (200 and 21 records), about 94 KB of JSON for 2010–2025 and regions 80–89. The bulk load iterates over `page` = 1 to `totalPages` and checks that the number of records equals `totalItems` and that no event id is repeated.

Run-ups. Run-up records (for example 6,429 for a single event) form a separate and much larger table. We do not load them: the indicator works at event level, and the events table already carries a per-tsunami summary (`maxWaterHeight` and `numRunups`). One page of run-ups for event 4682 is kept as evidence (`noaa_runups_event_4682_page1.json`); loading the full table is left as future work.

2.2 USGS FDSN Event Web Service

Owner	U.S. Geological Survey, Earthquake Hazards Program
URL	https://earthquake.usgs.gov/fdsnws/event/1/
Access method	REST API, CSV (also GeoJSON and QuakeML)
Authentication	None required
Rate limits	Maximum of 20,000 events per request (error 400 if exceeded) [2]. No numeric limit on requests per minute is documented; the documentation defines time × area limits that are not enforced yet but may be, and recommends the real-time feeds for automated applications. Team policy: one request per year, sequential, with caching.
License and attribution	U.S. public domain: most USGS information may be used without restriction. Credit is requested, not mandatory [3]; credit line used by the team: "USGS Earthquake Catalog data courtesy of the U.S. Geological Survey".
Fields used	`id`, `time`, `latitude`, `longitude`, `depth`, `mag`, `magType`, `net`, `place`, `status`, `updated`
Time coverage	Catalog of several decades; we load 2010–2025 with $M \geq 6$ inside the Pacific rectangle.
Update policy	Reviewed events are revised later (`updated`, magnitude and location can change). The service offers the `updatedafter` parameter, and deleted events return HTTP 409 unless `includedeleted` is requested; see Section 3.

Evidence. Query used for the historical sample; the unmodified response is stored in `workshop_1/samples/usgs_earthquakes_2010_2011_m6_pacific.csv`:

```
GET https://earthquake.usgs.gov/fdsnws/event/1/query?format=csv
    &starttime=2010-01-01&endtime=2012-01-01&minmagnitude=6
    &minlatitude=-60&maxlatitude=65&minlongitude=100&maxlongitude=300
```

The download script compares the number of downloaded records (343) with the result of the `count` method. A short excerpt of the response is in Appendix A.

Volume and partitioning. The `count` method gives 1,930 events for 2010–2025 (about 0.33 MB at roughly 169 bytes per row, estimated from the 343-row sample of 58 KB), as detailed in Table 2.

Year	Events	Year	Events
2010	157	2018	113
2011	186	2019	129
2012	110	2020	95
2013	115	2021	121
2014	130	2022	106
2015	121	2023	124
2016	127	2024	90
2017	86	2025	120

Table 2: USGS events per year ($M \geq 6$, Pacific rectangle), obtained with the `count` method on 8 November 2026. Total: 1,930.

Although these 1,930 events fit under the limit of 20,000 per request, we load one calendar year per request (16 requests). The documentation also defines time × area limits that are not enforced today but could be: a single 2010–2025 query over our rectangle amounts to about 146 million square-degree-days (5,844 days × 25,000 square degrees), above the 100 million cap of the `count` method, while each yearly partition amounts to about 9.15 million. Partitioning also allows retrying a single failed year without repeating the others and makes the load resumable. Each response is stored unchanged in the raw landing zone as `usgs_YYYYMMDD_HHMMSS.csv`. The rectangle is only a coarse query filter: it also contains areas outside the Pacific basin, such as the Caribbean and the Gulf of Mexico (longitudes 270 to 300) and parts of the Indian Ocean (south of Sumatra and Java, and off western Australia). Therefore a Pacific polygon filter is applied afterwards, during the load into the database, and the events it removes are counted in the `ingestion_run` record. The polygon is stored as GeoJSON in `config/pacific_polygon.geojson` and has the 25 vertices of Table 3. It follows the Pacific margins slightly inland (Alaska and British Columbia, the Americas, Australia and East Asia), extends south to latitude −60, and passes along the southern coasts of Java, Bali and Timor, so that the inner Indonesian seas (Java, Flores and Banda) are included while the Sunda trench on the Indian Ocean side is excluded, consistently with HazEL regions 80–89. Because the polygon crosses the antimeridian, its longitudes are stored as a continuous sequence from −255.5 to −66; before the point-in-polygon test, every event longitude $\lambda > 0$ is converted to $\lambda - 360$.

#	Latitude	Longitude	#	Latitude	Longitude
1	66.1°N	168.5°W	14	44.5°S	144°E
2	63.5°N	150°W	15	38°S	142°E
3	61.5°N	140°W	16	25°S	135°E
4	59.5°N	133°W	17	12°S	132°E
5	56°N	129°W	18	9.2°S	124°E
6	50°N	124°W	19	8.95°S	115°E
7	47°N	121°W	20	7.5°S	110°E
8	7.8°N	77.5°W	21	6°S	110°E
9	19.3°S	67.4°W	22	1.5°N	104.5°E
10	53.6°S	71°W	23	17.1°N	108°E
11	56°S	66°W	24	50°N	138.4°E
12	60°S	66°W	25	63°N	154.5°E
13	60°S	144°E			

Table 3: Vertices of the Pacific polygon, in order; the polygon closes from vertex 25 back to vertex 1.

Applied to the samples, the polygon removes 18 of the 343 USGS events (the 2010 Haiti earthquakes in the Caribbean, eight events on the Indian Ocean side of Sumatra, Java and Bali, one event in the Drake Passage, and seven inland earthquakes in South America and northeast Asia, five of them deeper than 500 km) and 7 of the 221 HazEL events. The seven HazEL events are landslide-generated waves in rivers, lakes and reservoirs far from the coast (in China, eastern Russia, the Yukon, British Columbia and Washington); they are outside the scope of a coastal tsunami monitor and are excluded on purpose.

3 Incremental Sources

3.1 USGS Real-Time GeoJSON Feed

URL	https://earthquake.usgs.gov/earthquakes/feed/v1.0/geojson.php
Feed used	Magnitude 4.5+, past 30 days
Format	GeoJSON
Update frequency	[TODO: the server sets a cache of 60 seconds]
Change detection	[TODO: `updated` field, stable `id`, Last-Modified header]

3.2 tsunami.gov Warning Centers

URL (NTWC)	https://www.tsunami.gov/events/xml/PAAQAtom.xml
URL (PTWC)	https://www.tsunami.gov/events/xml/PHEBAtom.xml
Format	Atom feed; each entry links to a CAP message
Content	[TODO: one latest message per feed; most messages are information statements]
Change detection	[TODO: entry id, `updated`, Last-Modified header]

3.3 Empirical Update Behavior

Two snapshots were taken on 7 November 2026 at 05:04 and 14:03 UTC (about 9 hours apart).

	USGS feed	tsunami.gov (NTWC, PTWC)
First snapshot (records)	714	1 per feed
Second snapshot (records)	726	1 per feed
New	22	0
Modified	25	0
Removed	10	0

Table 4: Changes between the two snapshots.

[TODO: Explain: all 10 removed events left the rolling 30-day window (none was deleted); fields that changed in the modified events (time, place, magnitude, station counts); event ids stayed stable.]

3.4 Revised or Deleted Records and Watermark Implications

[TODO: Watermark on `updated`, upsert on (source, id), content hash to separate real changes from timestamp-only updates, overlap window.]

4 Source Profiling and Data Quality

4.1 Profiling Summary

[TODO: Rows, null percentages, ranges, duplicated identifiers, timestamp format and time zone, coordinate validity and units for each source. Reference the notebook `notebooks/01_source_profiling.ipynb`.]

Source	Records	Duplicated ids	Notes
NOAA HazEL	[TODO:]	[TODO:]	[TODO:]
USGS FDSN	[TODO:]	[TODO:]	[TODO:]
USGS real-time	[TODO:]	[TODO:]	[TODO:]
tsunami.gov	[TODO:]	[TODO:]	[TODO:]

4.2 Data-Quality Problems

At least five, each with an example record.

#	Source	Problem	Example record
1	NOAA HazEL	Official warning status missing	[TODO:]
2	NOAA HazEL	Erroneous or doubtful events	[TODO:]
3	NOAA HazEL	Time of day or coordinates missing	[TODO:]
4	USGS FDSN	Columns that are always empty	[TODO:]
5	tsunami.gov	Location as free text; depth units differ	[TODO:]

4.3 Cross-Source Overlap

At least five candidate pairs describing the same real-world event.

#	Source A id	Source B id	Δt	Distance	Magnitude	What differs
1	PAAQ-1-to0j1g	nc75462637	-1.3 s	8.7 km	4.5 Ml / 4.96 mw	magnitude type, depth, location text
2	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
3	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
4	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
5	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]

5 Canonical Mapping

5.1 Field Mapping to the Canonical `event`

Canonical field	NOAA HazEL	USGS FDSN / feed	tsunami.gov (CAP)	Rule
`event_id`	[TODO:]	[TODO:]	[TODO:]	[TODO:]
`source`	[TODO:]	[TODO:]	[TODO:]	[TODO:]
`event_type`	[TODO:]	[TODO:]	[TODO:]	[TODO:]
`timestamp` (UTC)	[TODO:]	[TODO:]	[TODO:]	[TODO:]
`latitude`	[TODO:]	[TODO:]	[TODO:]	WGS84
`longitude`	[TODO:]	[TODO:]	[TODO:]	WGS84, -180 to 180
`severity`	[TODO:]	[TODO:]	[TODO:]	[TODO: value and unit/type]
`raw_reference`	[TODO:]	[TODO:]	[TODO:]	id or URL plus content hash

5.2 Domain Tables

[TODO: Proposed tables (for example `earthquake`, `tsunami_event`, `tsunami_alert`) with their attributes.]

5.3 Unit Conversions and Rules

[TODO: Timestamps (UTC), coordinates (WGS84), longitude normalization, depth units, magnitude types and severity scale.]

5.4 Draft ER Diagram



Figure 1: First draft of the entity-relationship diagram.

6 Ingestion Contract Draft

Source	Historical strategy	Incremental strategy	Duplicate key	Overlap window
NOAA HazEL	[TODO:]	[TODO:]	[TODO:]	[TODO:]
USGS FDSN	[TODO:]	[TODO:]	[TODO:]	[TODO:]
USGS real-time feed	–	[TODO:]	[TODO:]	[TODO:]
tsunami.gov	–	[TODO:]	[TODO:]	[TODO:]

6.1 Correctness After a Failed Run

[TODO: The watermark advances only after a successful commit; idempotent writes; what happens to a partially loaded batch.]

7 Project Definition

7.1 Project Overview

[TODO: Title, motivation, goal (data-engineering and analytical), scope, explicit exclusions and constraints.]

7.2 Functional User

[TODO: Role, profile, decisions supported, and why the dashboard is useful.]

7.3 User Stories

ID	Type	Story	MoSCoW
US1	Monitoring	As a [TODO: user], I want to [TODO: action] so that [TODO: benefit].	[TODO:]
US2	Monitoring	[TODO:]	[TODO:]
US3	Analysis	[TODO:]	[TODO:]
US4	Analysis	[TODO:]	[TODO:]
US5	Trust	[TODO:]	[TODO:]
US6	Trust	[TODO:]	[TODO:]

7.4 Indicator Proposal

[TODO: Name, inputs (magnitude, depth, location, historical tsunami association, official alert), rule, output scale, how it will be validated against historical events. State clearly that it is a monitoring aid and not a prediction, and list its limitations.]

7.5 Risks and Mitigation

Risk	Impact	Mitigation
Source outage	[TODO:]	[TODO:]
Rate limits	[TODO:]	[TODO:]
Schema change	[TODO:]	[TODO:]
License and attribution	[TODO:]	[TODO:]
Ambiguous identifiers	[TODO:]	[TODO:]

7.6 Architecture Overview

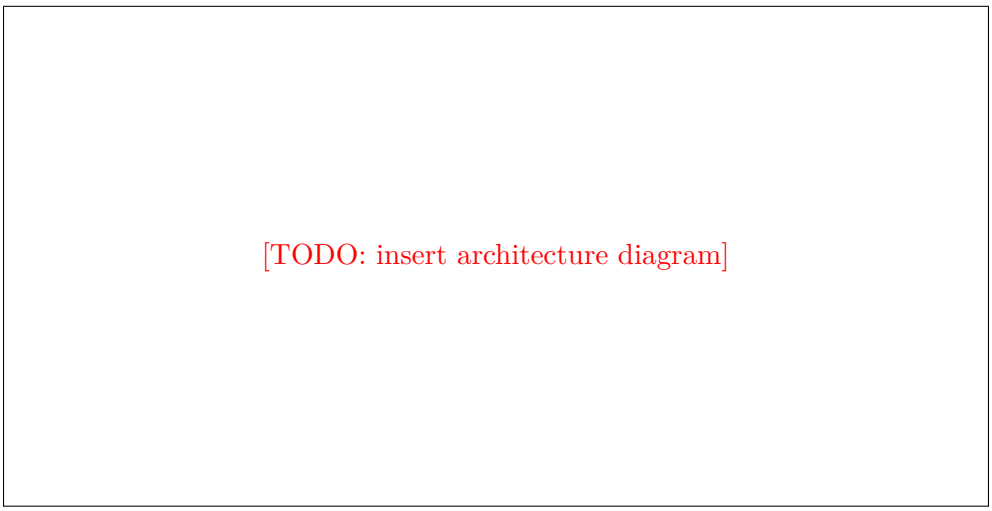


Figure 2: High-level architecture adapted to tsunami monitoring.

Conclusions and Open Questions

[TODO: Main decisions taken in this workshop and what remains open for Workshop 2.]

References

[1] National Geophysical Data Center / World Data Service. *NCEI/WDS Global Historical Tsunami Database*. NOAA National Centers for Environmental Information. doi:10.7289/V5PN93H7. Accessed through the HazEL API, https://www.ngdc.noaa.gov/hazel/view/swagger, 8 November 2026.

[2] U.S. Geological Survey. *API Documentation – Earthquake Catalog*. https://earthquake.usgs.gov/fdsnws/event/1/. Accessed 8 November 2026.

[3] U.S. Geological Survey. *Acknowledging or Crediting USGS*. https://www.usgs.gov/information-policies-and-instructions/acknowledging-or-crediting-usgs. Accessed 8 November 2026.

[4] U.S. Geological Survey. *Real-time Notifications, Feeds, and Web Services (GeoJSON summary feed)*. https://earthquake.usgs.gov/earthquakes/feed/v1.0/geojson.php. Accessed 7 November 2026.

[5] NOAA National Weather Service, Tsunami Warning Centers. *Tsunami event feeds*. https://www.tsunami.gov/. Accessed 7 November 2026.

A Real Response Snippets

Short excerpts of the unmodified responses stored in `workshop_1/samples/`. Each excerpt shows one record; "…" marks fields or lines left out for brevity.

A.1 NOAA HazEL – tsunami events

File `noaa_tsunamis_pacific_2010_2025.json`, record of the 2011 Tohoku tsunami, and the pagination fields of the response:

```
{"items": [ ...
  {"id": 5413, "year": 2011, "month": 3, "day": 11, "hour": 5, "minute": 46,
   "latitude": 38.297, "longitude": 142.373, "causeCode": 1,
   "eventValidity": 4, "eqMagnitude": 9.1, "eqDepth": 29,
   "maxWaterHeight": 39.26, "warningStatusId": null, "regionCode": 85,
   "numRunups": 6429, "country": "JAPAN", "locationName": "HONSHU ISLAND", ...},
  ... ],
 "page": 1, "totalPages": 2, "itemsPerPage": 200, "totalItems": 221}
```

A.2 USGS FDSN – earthquake catalog

File `usgs_earthquakes_2010_2011_m6_pacific.csv`, header and the row of the same earthquake:

```
time,latitude,longitude,depth,mag,magType,nst,gap,dmin,rms,net,id,updated,place,type,
    horizontalError,depthError,magError,magNst,status,locationSource,magSource
2011-03-11T05:46:24.120Z,38.297,142.373,29,9.1,mww,541,9.5,,1.16,official,
    official20110311054624120_30,2026-10-16T10:38:09.160Z,"2011 Great Tohoku Earthquake, Japan",
    earthquake,,,,,reviewed,us,official
```

The two records describe the same event: same origin minute, epicenter and magnitude, and the HazEL `eqDepth` (29) equals the USGS depth in km.

A.3 USGS real-time feed

File `usgs_realtime_20261107_050434.geojson`, feed metadata and first feature (times in epoch milliseconds, coordinates as longitude, latitude, depth in km):

```
{"type": "FeatureCollection",
 "metadata": {"generated": 1794027872000,
   "url": "https://earthquake.usgs.gov/earthquakes/feed/v1.0/summary/4.5_month.geojson",
   "title": "USGS Magnitude 4.5+ Earthquakes, Past Month", "status": 200, "count": 714, ...},
 "features": [
  {"type": "Feature", "id": "us7000twmr",
   "properties": {"mag": 4.8, "place": "36 km WNW of Poso, Indonesia",
     "time": 1793992021269, "updated": 1794002237040, "tsunami": 0,
     "magType": "mb", "status": "reviewed", "type": "earthquake", ...},
   "geometry": {"type": "Point", "coordinates": [120.4589, -1.2393, 10]}},
  ... ]}
```

A.4 tsunami.gov – PTWC Atom feed

File `ptwc_20261107_050434.xml`, the single entry of the feed (an information statement); the linked CAP message is stored in `ptwc_cap1_20261107_050434.xml`:

```
<entry>
  <title>BONIN ISLANDS JAPAN REGION</title><updated>2026-10-25T02:27:01Z</updated>
  <geo:lat>28.</geo:lat>
  <geo:long>139.9</geo:long>
```

```
  <summary type="xhtml"> ... <strong>Category:</strong> Information<br/>
    <strong>Bulletin Issue Time: </strong> 2026.10.25 02:27:01 UTC <br/>
    <strong>Preliminary Magnitude: </strong>7.0(Mwp)<br/> ... </summary>
  <id>urn:uuid:ba64bc82-07f7-45fb-ab6c-25420fca6851</id>
  <link rel="related" title="CapXML document" type="application/cap+xml"
        href="https://www.tsunami.gov/events/PHEB/2026/10/24/26297050/1/WEGM42/PHEBCAP.xml" />
</entry>
```